

US Interstate Wage Gaps, 2004–2024

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Research question. Did the U.S. interstate wage gap widen between 2004 and 2024 — and is the answer different at the state and county levels?

Why a manager at EPI should care. EPI allocates research and advocacy resources across federal wage-policy campaigns (the Raise the Wage Act, federal labor-law reform) and state-level work through its EARN network. The headline answer to whether geographic wage gaps are widening shapes both: a clear widening at the state level argues for federal action and place-based investment; broad divergence at the county level argues for state and metropolitan policy; stability at either scale argues against “geographic divergence” framings and toward industry-specific or individual-level policy instead. The answer also informs the geography chapter of EPI’s flagship *State of Working America* report.

Two stylized facts that motivate the question.

1. **The gap is large in level terms.** In 2024 the highest-wage state (Massachusetts) paid an average annual wage of about \$96,000 — roughly 1.89 times the lowest (Mississippi, about \$51,000). A gap of that size in the level naturally raises the question of whether it has been growing in proportional terms.
2. **The literature finds that interstate convergence stalled, but is ambiguous about whether it has reversed.** Ganong and Shoag (2017) document that the long-run convergence of state per-capita incomes — roughly 2% per year from 1880 to 1980 (Barro and Sala-i-Martin, 1992) — effectively ended around 1980, largely because rising housing costs in rich states eliminated the financial benefit of migration. Whether “stopped converging” has turned into “started diverging” is an open empirical question on the 2004–2024 window.

```
library(tidyverse)
library(scales)
library(knitr)
library(ggrepel)

theme_set(theme_minimal(base_size = 11))
```

Section 1: Raw Data Description

Our raw data is the BLS Quarterly Census of Employment and Wages (QCEW), the federal payroll census underlying every official state and county wage statistic. The 2004–2024 annual-by-area files cover roughly 93,000 area-year records spanning every U.S. state and county over 21 years, with the underlying microdata covering about 140 million workers and 10 million establishments per year — approximately 95% of all U.S. wage and salary employment.

Each year is delivered as a folder of ~4,400 CSVs, one per area (state, county, MSA, or “Unknown Or Undefined”), and within each CSV rows are broken out by industry $\times$ ownership sector $\times$ aggregation level. Key fields we retain are:

Field	Description
area_fips	5-digit FIPS code (06000 = California statewide; 06037 = LA County)
area_title	State or county name
year	Calendar year, 2004–2024
avg_annual_pay	Average annual wages per worker, nominal USD
annual_avg_emplvl	Average annual employment level (workers)
total_annual_wages	Total wages paid, nominal USD

The **dependent variable** is `avg_annual_pay`. Key **independent variables** span three dimensions:

- **Time** — `year`, 2004–2024, the dimension along which we test whether the gap widens.
- **Cross-sectional identity** — `area_fips`, identifying the 50 U.S. states (Section 2–4) and ~3,100 counties (Section 5). DC, Puerto Rico, and the Virgin Islands are excluded from the 50-state framing.
- **State/county scale** — `annual_avg_emplvl`, used as context and as the minimum-employment threshold for the county sample (`emplvl` 500).

```
# Replace the URL below with the Dropbox link to qcew_state_panel.csv.
# Make sure the URL ends in ?dl=1 (not ?dl=0).
panel_url <- "https://www.dropbox.com/scl/fi/ztkjy06gxm9f0pbaugp67/qcew_state_panel.csv?rlkey="

panel_raw <- read_csv(
  panel_url,
  show_col_types = FALSE,
  # Force area_fips to character so leading zeros (e.g. "01000") survive.
  col_types = cols(area_fips = col_character(), .default = col_guess())
)

dim(panel_raw)
```

```
[1] 1113  11
```

```

panel_raw |>
  filter(agglvl_code == 50, industry_code == 10, own_code == 0) |>
  filter(!area_fips %in% c("11000", "72000", "78000")) |>
  summarise(across(
    c(avg_annual_pay, annual_avg_emplvl, year),
    list(Mean = mean, SD = sd, Min = min, Median = median, Max = max),
    .names = "{.col}__{.fn}"
  )) |>
  pivot_longer(everything(), names_sep = "__",
               names_to = c("Variable", "Statistic")) |>
  pivot_wider(names_from = Statistic, values_from = value) |>
  mutate(across(c(Mean, SD, Min, Median, Max), \(x) round(x, 0))) |>
  kable(format.args = list(big.mark = ","))

```

Table 1: Summary statistics, analytical state-year extract (50 states x 21 years = 1,050 state-years, 2004–2024). avg_annual_pay is nominal USD.

Variable	Mean	SD	Min	Median	Max
avg_annual_pay	49,574	12,106	27,830	47,213	96,109
annual_avg_emplvl	2,760,314	3,008,208	248,051	1,843,066	18,183,696
year	2,014	6	2,004	2,014	2,024

Data quality flags.

1. **avg_annual_pay is nominal.** We deflate it to 2024 dollars using BLS CPI-U annual averages (series CUUR0000SA0, joined on year) before any cross-year comparison. Without deflation a “wage rose 60%” pattern is mostly inflation.
2. **Coverage is UI-employees only.** QCEW excludes the unincorporated self-employed, gig workers, unpaid family workers, and railroad workers. Our claim is about formal-payroll wages, not all labor earnings.
3. **DC is a structural outlier.** Federal-employee concentration pushes DC’s average wage roughly \$25K above any state in every year. We exclude DC from the main 50-state analysis and re-include it as a robustness check (Section 4, Check 2).
4. **County coverage is dense but not universal.** A small number of county-years (~9 out of 67,000) are suppressed for confidentiality; we drop suppressed and zero-wage rows in Section 5.

Section 2: Sample Construction

We perform three transformations on the raw panel: (1) select state-total rows, (2) restrict to the 50 US states, and (3) deflate nominal wages to 2024 dollars using BLS CPI-U.

Filter to state-total rows

The QCEW source files contain rows at many aggregation levels. We keep only the single fully-collapsed state-total row in each statewide file, identified by `agglvl_code == 50` (State, Total Covered), `industry_code == 10` (Total, all industries), and `own_code == 0` (Total Covered). The input CSV was pre-filtered to these rows, but we re-apply the filter visibly to document the criterion.

```
panel_state_total <- panel_raw |>
  filter(agglvl_code == 50, industry_code == 10, own_code == 0) |>
  select(-agglvl_code, -industry_code, -own_code)

nrow(panel_state_total)
```

```
[1] 1113
```

Drop non-state jurisdictions

DC is a federal enclave whose economy is structurally different from the 50 states; Puerto Rico and the Virgin Islands sit outside the 50-state framing of our research question. We drop all three by FIPS code.

```
non_states <- c("11000", "72000", "78000") # DC, PR, VI

state_panel <- panel_state_total |>
  filter(!area_fips %in% non_states) |>
  mutate(state = str_remove(area_title, " -- Statewide"))
```

Add Census region

```
state_panel <- state_panel |>
  mutate(
    state_fips_2 = substr(area_fips, 1, 2),
    region = case_when(
      state_fips_2 %in% c("09", "23", "25", "33", "34", "36", "42", "44", "50")
        ~ "Northeast",
      state_fips_2 %in% c("17", "18", "19", "20", "26", "27", "29", "31", "38", "39",
        "46", "55") ~ "Midwest",
      state_fips_2 %in% c("01", "05", "10", "12", "13", "21", "22", "24", "28", "37",
        "40", "45", "47", "48", "51", "54") ~ "South",
      state_fips_2 %in% c("02", "04", "06", "08", "15", "16", "30", "32", "35", "41",
        "49", "53", "56") ~ "West"
    )
  ) |>
  select(-state_fips_2)
```

Deflate to 2024 dollars

`wage_real` expresses every year's wage in 2024 purchasing power. The 2024 row is unchanged; every earlier year is scaled up by cumulative inflation through 2024.

```

cpi_2024 <- state_panel |>
  filter(year == 2024) |>
  pull(cpi_u) |>
  first()

state_panel <- state_panel |>
  mutate(wage_real = avg_annual_pay * (cpi_2024 / cpi_u))

cpi_2024
```

```
[1] 313.689
```

Row-count accounting

```

tibble(
  step = c(
    "Raw panel (53 jurisdictions x 21 years)",
    "After state-total filter",
    "After dropping DC, PR, VI",
    "Expected (50 states x 21 years)"
  ),
  rows = c(
    nrow(panel_raw),
    nrow(panel_state_total),
    nrow(state_panel),
    50 * 21
  )
) |>
  kable()
```

step	rows
Raw panel (53 jurisdictions x 21 years)	1113
After state-total filter	1113
After dropping DC, PR, VI	1050
Expected (50 states x 21 years)	1050

The analytical panel `state_panel` has 1050 rows (50 states $\times$ 21 years), 11 columns, and serves as the input to every analysis below.

Section 3: Analysis

Analysis 1 — Dispersion ratios over time (Figure 1)

For each year, we compute the 10th, 25th, 50th, 75th, and 90th percentiles of `wage_real` across the 50 states, then form three dispersion ratios: P90/P10 (the headline gap), P90/P50 (upper-half dispersion), and P50/P10 (lower-half dispersion). A rising P90/P10 line means the interstate gap is widening; whether the rise sits in P90/P50 or P50/P10 tells us which half is doing the work.

```
percentiles_year <- state_panel |>
  group_by(year) |>
  summarise(
    p10 = quantile(wage_real, 0.10),
    p25 = quantile(wage_real, 0.25),
    p50 = quantile(wage_real, 0.50),
    p75 = quantile(wage_real, 0.75),
    p90 = quantile(wage_real, 0.90),
    .groups = "drop"
  ) |>
  mutate(
    `P90/P10` = p90 / p10,
    `P90/P50` = p90 / p50,
    `P50/P10` = p50 / p10
  )
```

```
ratios_long <- percentiles_year |>
  select(year, `P90/P10`, `P90/P50`, `P50/P10`) |>
  pivot_longer(-year, names_to = "ratio", values_to = "value") |>
  mutate(ratio = factor(ratio, levels = c("P90/P10", "P90/P50", "P50/P10")))

ratio_palette <- c("P90/P10" = "#1f2a44",
                  "P90/P50" = "#992f31",
                  "P50/P10" = "#7c8a9b")

ratio_widths <- c("P90/P10" = 1.4,
                  "P90/P50" = 0.8,
                  "P50/P10" = 0.8)

ggplot(ratios_long, aes(x = year, y = value,
                       color = ratio, linewidth = ratio)) +
  annotate("rect", xmin = 2008, xmax = 2009, ymin = -Inf, ymax = Inf,
         fill = "gray85", alpha = 0.5) +
  annotate("rect", xmin = 2020, xmax = 2021, ymin = -Inf, ymax = Inf,
         fill = "gray85", alpha = 0.5) +
  annotate("text", x = 2008.5, y = Inf, label = "Recession",
         vjust = 1.5, size = 3, family = "serif", color = "gray40") +
  annotate("text", x = 2020.5, y = Inf, label = "COVID",
         vjust = 1.5, size = 3, family = "serif", color = "gray40") +
```

```

geom_line() +
scale_color_manual(values = ratio_palette) +
scale_linewidth_manual(values = ratio_widths) +
scale_x_continuous(breaks = seq(2004, 2024, 4)) +
guides(linewidth = "none") +
labs(x = NULL, y = "Ratio (across 50 states)", color = NULL) +
theme(
  text = element_text(family = "serif"),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray92"),
  legend.position = "inside",
  legend.position.inside = c(0.15, 0.85),
  legend.background = element_rect(fill = "white", color = NA)
)

```

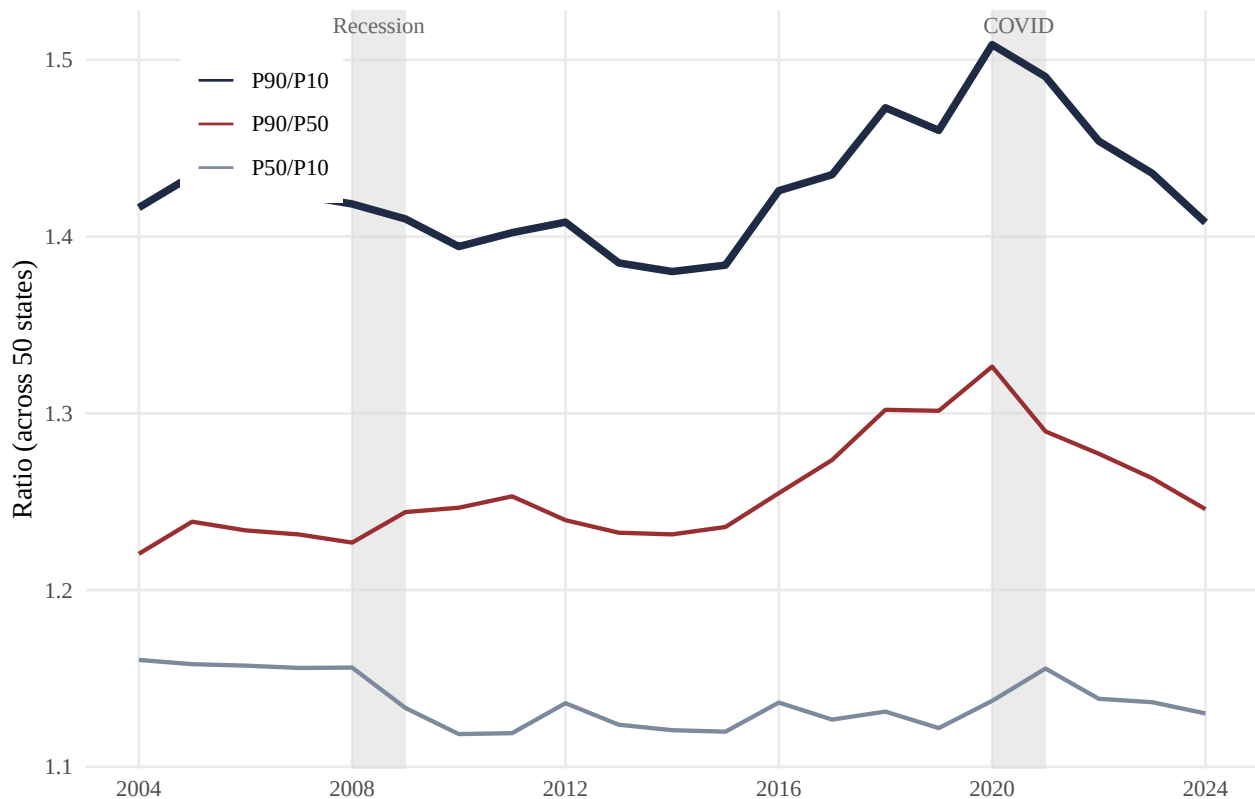


Figure 1: Figure 1: Three dispersion ratios across the 50 US states, 2004–2024. P90/P10 (bold) is the headline interstate gap. Shaded bands flag the 2008–2009 recession and 2020–2021 COVID years.

```

percentiles_year |>
  filter(year %in% c(2004, 2014, 2024)) |>
  select(year, `P90/P10`, `P90/P50`, `P50/P10`) |>
  mutate(across(c(`P90/P10`, `P90/P50`, `P50/P10`), \(x) round(x, 3))) |>
  kable()

```

Table 2: Table A1: Dispersion ratios for selected years.

year	P90/P10	P90/P50	P50/P10
2004	1.416	1.220	1.160
2014	1.380	1.232	1.121
2024	1.408	1.246	1.130

Analysis 2 — Indexed percentile trajectories (Figure 2)

Analysis 1 reports ratios; this view normalizes each percentile to its own 2004 value (=100) so we can see whether one part of the distribution broke away from the rest. If all five lines move together, there is no divergence. If P90 lifts above P10–P75 that cluster, a small set of high-wage states is pulling away. If all five fan out symmetrically, divergence is broad.

```
# Reshape percentiles_year to long form, index each percentile to its 2004 value.
indexed_long <- percentiles_year |>
  select(year, p10, p25, p50, p75, p90) |>
  pivot_longer(~year, names_to = "percentile", values_to = "value") |>
  group_by(percentile) |>
  mutate(index = value / value[year == 2004] * 100) |>
  ungroup() |>
  mutate(
    percentile = factor(
      percentile,
      levels = c("p90", "p75", "p50", "p25", "p10"),
      labels = c("P90", "P75", "P50", "P25", "P10")
    )
  )

pct_palette <- c(
  "P90" = "#992f31",
  "P75" = "#1f2a44",
  "P50" = "#4a5a7a",
  "P25" = "#7c8a9b",
  "P10" = "#b1bdc8"
)

pct_widths <- c("P90" = 1.4, "P75" = 0.8, "P50" = 0.8, "P25" = 0.8, "P10" = 0.8)

end_labels <- indexed_long |> filter(year == 2024)

ggplot(indexed_long, aes(x = year, y = index,
                        color = percentile, linewidth = percentile)) +
  geom_hline(yintercept = 100, color = "gray70", linetype = "dashed",
            linewidth = 0.4) +
  geom_line() +
  geom_text(
```



```

data = end_labels,
aes(label = percentile),
hjust = -0.2, size = 3.2, family = "serif", show.legend = FALSE
) +
scale_color_manual(values = pct_palette) +
scale_linewidth_manual(values = pct_widths) +
scale_x_continuous(breaks = seq(2004, 2024, 4),
                    limits = c(2004, 2025.5)) +
guides(linewidth = "none", color = "none") +
labs(x = NULL, y = "Indexed value (2004 = 100)") +
theme(
  text = element_text(family = "serif"),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray92")
)

```

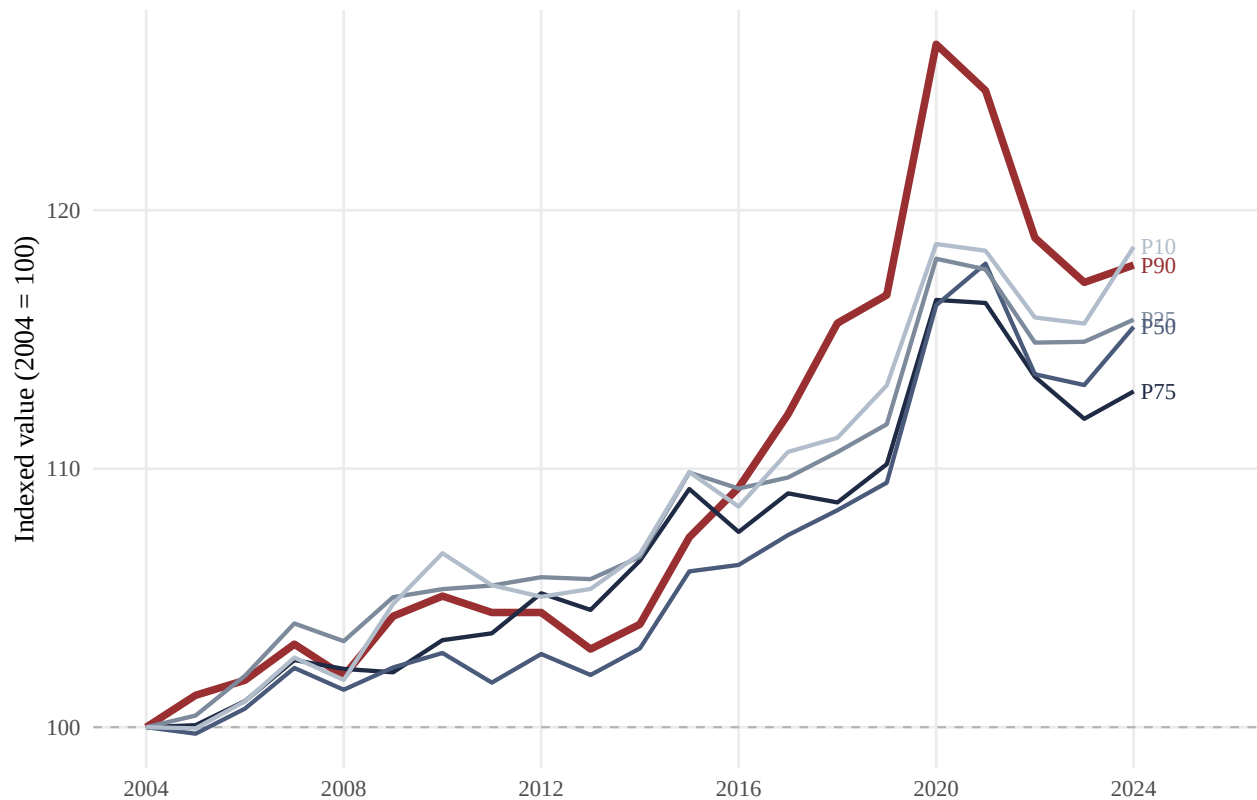


Figure 2: Figure 2: Across-state percentiles of real wages, indexed to 100 in 2004.

```

indexed_long |>
  filter(year %in% c(2004, 2014, 2024)) |>
  select(year, percentile, index) |>
  mutate(index = round(index, 1)) |>
  pivot_wider(names_from = percentile, values_from = index) |>
  kable()

```

Table 3: Table A2: Indexed percentile values for selected years (2004 = 100).

year	P10	P25	P50	P75	P90
2004	100.0	100.0	100.0	100.0	100.0
2014	106.7	106.6	103.1	106.4	104.0
2024	118.6	115.8	115.5	113.0	117.9

Analysis 3 — Growth-incidence scatter (Figure 3)

Analyses 1 and 2 describe the distribution as a whole. This view attaches names to numbers: per-state real wage growth from 2004 to 2024 plotted against each state's 2004 wage. A flat scatter means every state grew at about the same rate (no rich-getting-richer pattern). A positively-sloped LOESS line means already-rich states grew faster — the “top pulling away” mechanism. A negatively-sloped LOESS line means convergence (catch-up growth).

```
growth_by_state <- state_panel |>
  filter(year %in% c(2004, 2024)) |>
  select(state, region, year, wage_real) |>
  pivot_wider(names_from = year, values_from = wage_real,
              names_prefix = "wage_real_") |>
  mutate(pct_growth_2004_2024 = (wage_real_2024 / wage_real_2004) - 1)

top_5 <- growth_by_state |> slice_max(pct_growth_2004_2024, n = 5)
bottom_5 <- growth_by_state |> slice_min(pct_growth_2004_2024, n = 5)
to_label <- bind_rows(top_5, bottom_5)
```

```
region_palette <- c(
  "Northeast" = "#1f2a44",
  "Midwest" = "#4a7a4a",
  "South" = "#992f31",
  "West" = "#b58b3a"
)

ggplot(growth_by_state,
  aes(x = wage_real_2004 / 1000, y = pct_growth_2004_2024)) +
  geom_smooth(method = "loess", se = FALSE,
             color = "gray40", linewidth = 0.6, linetype = "dashed",
             formula = y ~ x) +
  geom_point(aes(color = region), size = 2.4, alpha = 0.9) +
  ggrepel::geom_text_repel(
    data = to_label,
    aes(label = state),
    size = 3.0, family = "serif",
    box.padding = 0.4, max.overlaps = Inf, seed = 42
  ) +
  scale_color_manual(values = region_palette) +
```

```

scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
scale_x_continuous(labels = scales::label_dollar(suffix = "K")) +
labs(
  x = "2004 real wage (thousands of 2024 dollars)",
  y = "Cumulative real wage growth, 2004 → 2024",
  color = NULL
) +
theme(
  text = element_text(family = "serif"),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray92"),
  legend.position = "top"
)

```

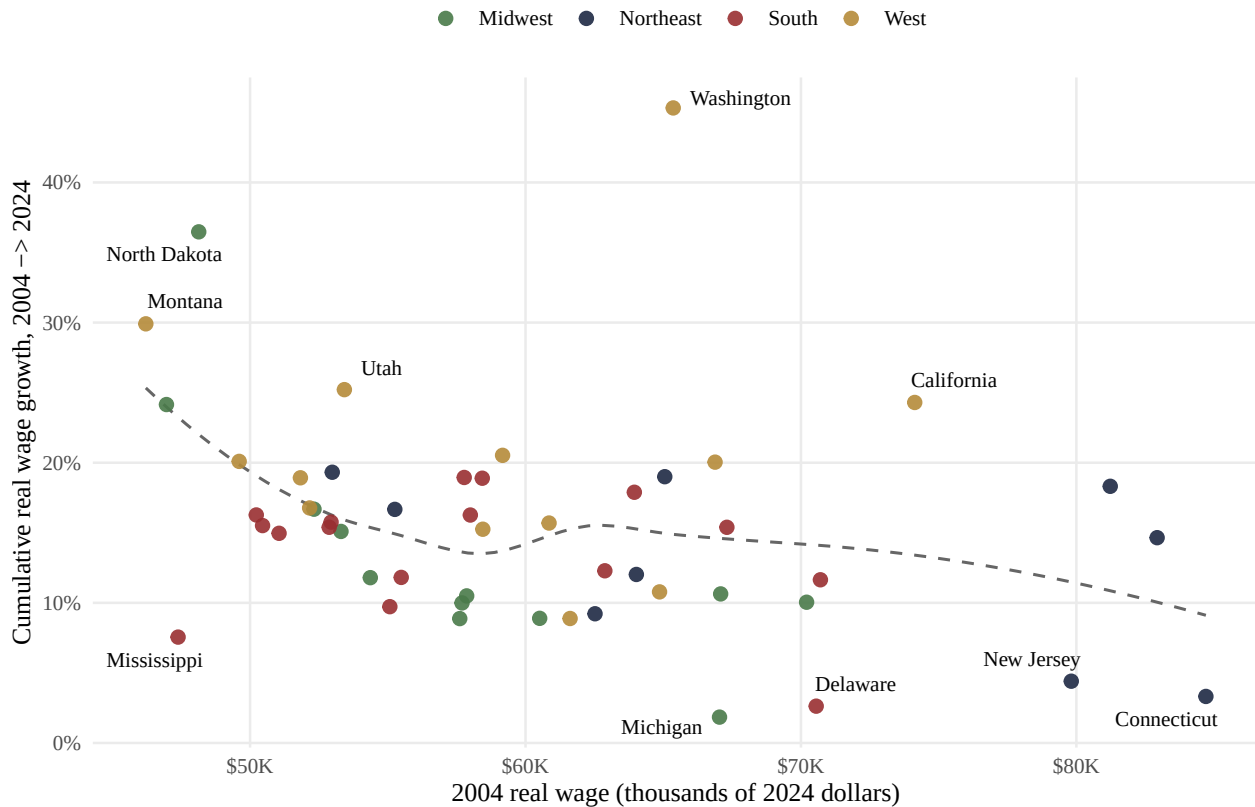


Figure 3: Figure 3: 2004 real wage vs. 2004–2024 real wage growth, by state. Five highest- and five lowest-growth states labeled. LOESS trend overlaid.

```

bind_rows(
  top_5 |> mutate(rank_group = "Top 5 (fastest growth)"),
  bottom_5 |> mutate(rank_group = "Bottom 5 (slowest growth)")
) |>
mutate(
  `2004 real wage` = scales::dollar(round(wage_real_2004)),
  `2024 real wage` = scales::dollar(round(wage_real_2024)),

```

```

  `Real growth` = scales::percent(pct_growth_2004_2024, accuracy = 0.1)
) |>
select(rank_group, state, region, `2004 real wage`, `2024 real wage`,
  `Real growth`) |>
kable()

```

Table 4: Table A3: Five fastest- and five slowest-growing states, 2004–2024.

rank_group	state	region	2004 real wage	2024 real wage	Real growth
Top 5 (fastest growth)	Washington	West	\$65,363	\$94,985	45.3%
Top 5 (fastest growth)	North Dakota	Midwest	\$48,136	\$65,694	36.5%
Top 5 (fastest growth)	Montana	West	\$46,215	\$60,038	29.9%
Top 5 (fastest growth)	Utah	West	\$53,423	\$66,894	25.2%
Top 5 (fastest growth)	California	West	\$74,131	\$92,145	24.3%
Bottom 5 (slowest growth)	Michigan	Midwest	\$67,044	\$68,281	1.8%
Bottom 5 (slowest growth)	Delaware	South	\$70,554	\$72,408	2.6%
Bottom 5 (slowest growth)	Connecticut	North-east	\$84,703	\$87,516	3.3%
Bottom 5 (slowest growth)	New Jersey	North-east	\$79,816	\$83,334	4.4%
Bottom 5 (slowest growth)	Mississippi	South	\$47,385	\$50,966	7.6%

Analysis 4 — How concentrated is any movement? (Figure 4)

Analysis 1 showed the P90/P10 ratio was roughly stable over 21 years. Analysis 3 showed real growth-rate heterogeneity at the state level (1.8% to 45.3%). Analysis 4 asks: how much of even the small movement in the headline ratio is mechanically driven by the handful of extreme-growth states? We recompute P90/P10 dropping (a) the top 5 fastest-growing states and (b) the bottom 5 slowest-growing states, and plot all three series on the same axes.

In the plan’s hypothesized world (top pulling away), dropping the top 5 would flatten the trend dramatically. In a world of broad divergence, neither drop would matter much. In our actual data the headline trend is already flat, so we are testing whether even that small movement is extreme-state-driven.

```

compute_p90p10 <- function(df) {
  df |>
  group_by(year) |>
  summarise(
    p10 = quantile(wage_real, 0.10),
    p90 = quantile(wage_real, 0.90),

```

```

    ratio = p90 / p10,
    .groups = "drop"
  )
}

top_5_states    <- top_5$state
bottom_5_states <- bottom_5$state

ratio_full <- compute_p90p10(state_panel) |>
  mutate(series = "Full sample (50 states)")

ratio_no_top5 <- state_panel |>
  filter(!state %in% top_5_states) |>
  compute_p90p10() |>
  mutate(series = "Drop 5 fastest-growth states")

ratio_no_bot5 <- state_panel |>
  filter(!state %in% bottom_5_states) |>
  compute_p90p10() |>
  mutate(series = "Drop 5 slowest-growth states")

ratios_all <- bind_rows(ratio_full, ratio_no_top5, ratio_no_bot5) |>
  mutate(series = factor(
    series,
    levels = c("Full sample (50 states)",
               "Drop 5 fastest-growth states",
               "Drop 5 slowest-growth states")
  ))

```

```

series_palette <- c(
  "Full sample (50 states)"      = "#1f2a44",
  "Drop 5 fastest-growth states" = "#992f31",
  "Drop 5 slowest-growth states" = "#7c8a9b"
)

series_lty <- c(
  "Full sample (50 states)"      = "solid",
  "Drop 5 fastest-growth states" = "dashed",
  "Drop 5 slowest-growth states" = "dotdash"
)

series_lwd <- c(
  "Full sample (50 states)"      = 1.3,
  "Drop 5 fastest-growth states" = 0.9,
  "Drop 5 slowest-growth states" = 0.9
)

ggplot(ratios_all,
  aes(x = year, y = ratio,

```

```

    color = series, linetype = series, linewidth = series)) +
  geom_line() +
  scale_color_manual(values = series_palette) +
  scale_linetype_manual(values = series_lty) +
  scale_linewidth_manual(values = series_lwd) +
  scale_x_continuous(breaks = seq(2004, 2024, 4)) +
  guides(linewidth = "none", linetype = guide_legend(override.aes = list(linewidth = 0.8))) +
  labs(x = NULL, y = "P90/P10 ratio (across states)",
    color = NULL, linetype = NULL) +
  theme(
    text = element_text(family = "serif"),
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "gray92"),
    legend.position = "top",
    legend.text = element_text(size = 9)
  )

```

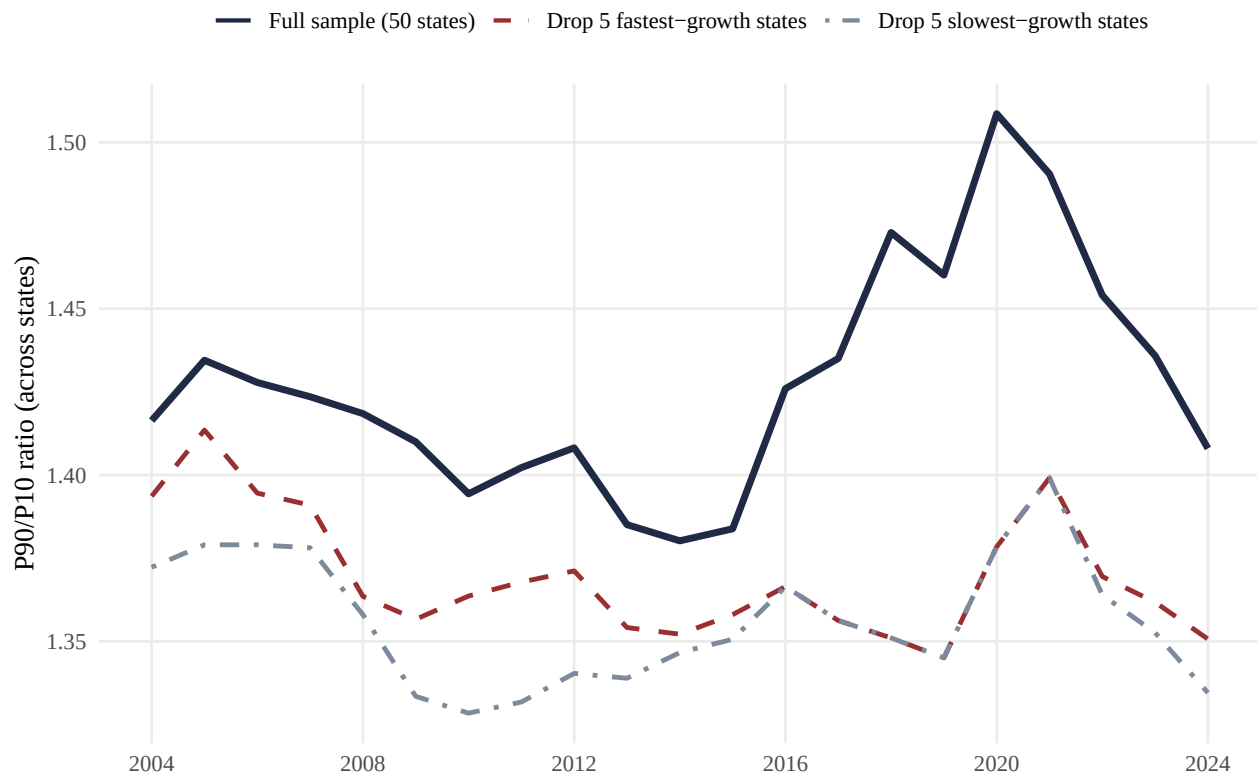


Figure 4: Figure 4: P90/P10 ratio across states, recomputed three ways. The trimmed series (dashed) sit consistently below the full sample by 0.03–0.07 and trend slightly downward (1.37 → 1.34), while the full sample is flat (1.42 → 1.41). Headline stability is sustained by extreme states at both tails: the middle 40 states converged slightly over 2004–2024.

```

ratios_all |>
  filter(year %in% c(2004, 2014, 2024)) |>
  select(series, year, ratio) |>

```

```
mutate(ratio = round(ratio, 3)) |>
pivot_wider(names_from = year, values_from = ratio) |>
kable()
```

Table 5: Table A4: P90/P10 ratio for selected years, three samples.

series	2004	2014	2024
Full sample (50 states)	1.416	1.380	1.408
Drop 5 fastest-growth states	1.394	1.352	1.351
Drop 5 slowest-growth states	1.372	1.347	1.335

Section 4: Robustness Checks

The state-level finding (Analyses 1–4) needs two stress tests: one against the COVID composition anomaly visible in every figure, and one against the decision to exclude DC.

Check 1 — COVID exclusion

The COVID period (2020–2021) created a mechanical spike in average wages because low-wage workers were disproportionately laid off. We re-run the headline P90/P10 trend dropping those two years to confirm the residual COVID effect does not drive the 2024 endpoint.

```
ratio_no_covid <- state_panel |>
  filter(!year %in% c(2020, 2021)) |>
  compute_p90p10() |>
  mutate(series = "Drop 2020 and 2021")

ratio_full_for_compare <- ratio_full |>
  mutate(series = "Full sample (all years)")

ratios_covid_check <- bind_rows(ratio_full_for_compare, ratio_no_covid) |>
  mutate(series = factor(
    series,
    levels = c("Full sample (all years)", "Drop 2020 and 2021")
  ))
```

```
ggplot(ratios_covid_check,
  aes(x = year, y = ratio, color = series, linetype = series)) +
  annotate("rect", xmin = 2020, xmax = 2021, ymin = -Inf, ymax = Inf,
    fill = "gray85", alpha = 0.4) +
  geom_line(linewidth = 1.0) +
  scale_color_manual(values = c("Full sample (all years)" = "#1f2a44",
    "Drop 2020 and 2021" = "#992f31")) +
```

```

scale_linetype_manual(values = c("Full sample (all years)" = "solid",
                                "Drop 2020 and 2021"      = "dashed")) +
scale_x_continuous(breaks = seq(2004, 2024, 4)) +
labs(x = NULL, y = "P90/P10 ratio (across 50 states)",
     color = NULL, linetype = NULL) +
theme(
  text = element_text(family = "serif"),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray92"),
  legend.position = "top"
)

```

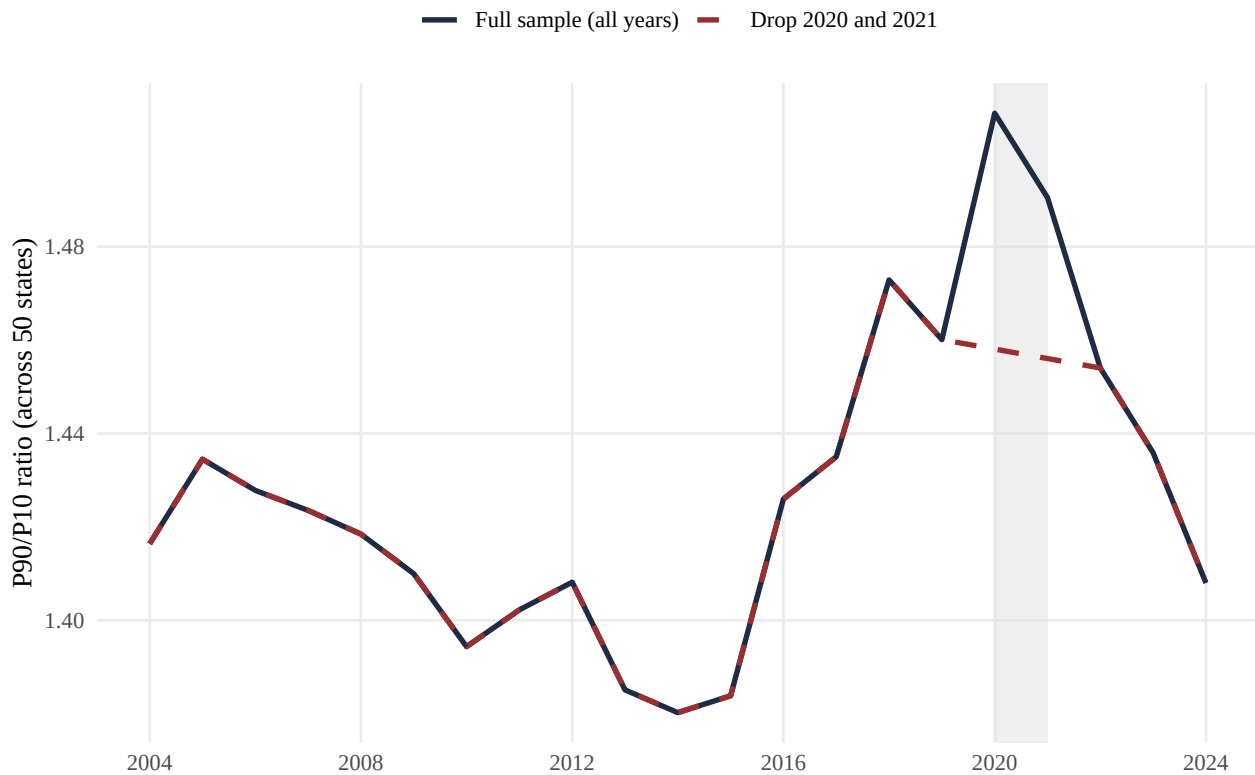


Figure 5: Robustness Check 1: P90/P10 with and without 2020–2021. The post-COVID trend without those two years returns to roughly the 2004 baseline, confirming the headline stability is not an artifact of pandemic composition effects.

Check 2 — DC sensitivity

DC is structurally distinct (high federal-employee concentration, no governance counterpart among the 50 states), but excluding it is a choice that could shape the headline. We re-run the P90/P10 trend with DC re-included.

```

state_panel_with_dc <- panel_raw |>
  filter(agglvl_code == 50, industry_code == 10, own_code == 0) |>

```



```

filter(!area_fips %in% c("72000", "78000")) |> # only drop PR, VI
mutate(state = str_remove(area_title, " -- Statewide")) |>
mutate(wage_real = avg_annual_pay * (cpi_2024 / cpi_u))

ratio_with_dc <- compute_p90p10(state_panel_with_dc) |>
  mutate(series = "50 states + DC")

ratio_dc_check <- bind_rows(
  ratio_full |> mutate(series = "50 states (main)"),
  ratio_with_dc
) |>
  mutate(series = factor(series,
    levels = c("50 states (main)", "50 states + DC")))

ggplot(ratio_dc_check,
  aes(x = year, y = ratio, color = series, linetype = series)) +
  geom_line(linewidth = 1.0) +
  scale_color_manual(values = c("50 states (main)" = "#1f2a44",
    "50 states + DC" = "#992f31")) +
  scale_linetype_manual(values = c("50 states (main)" = "solid",
    "50 states + DC" = "dashed")) +
  scale_x_continuous(breaks = seq(2004, 2024, 4)) +
  labs(x = NULL, y = "P90/P10 ratio",
    color = NULL, linetype = NULL) +
  theme(
    text = element_text(family = "serif"),
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "gray92"),
    legend.position = "top"
  )

```

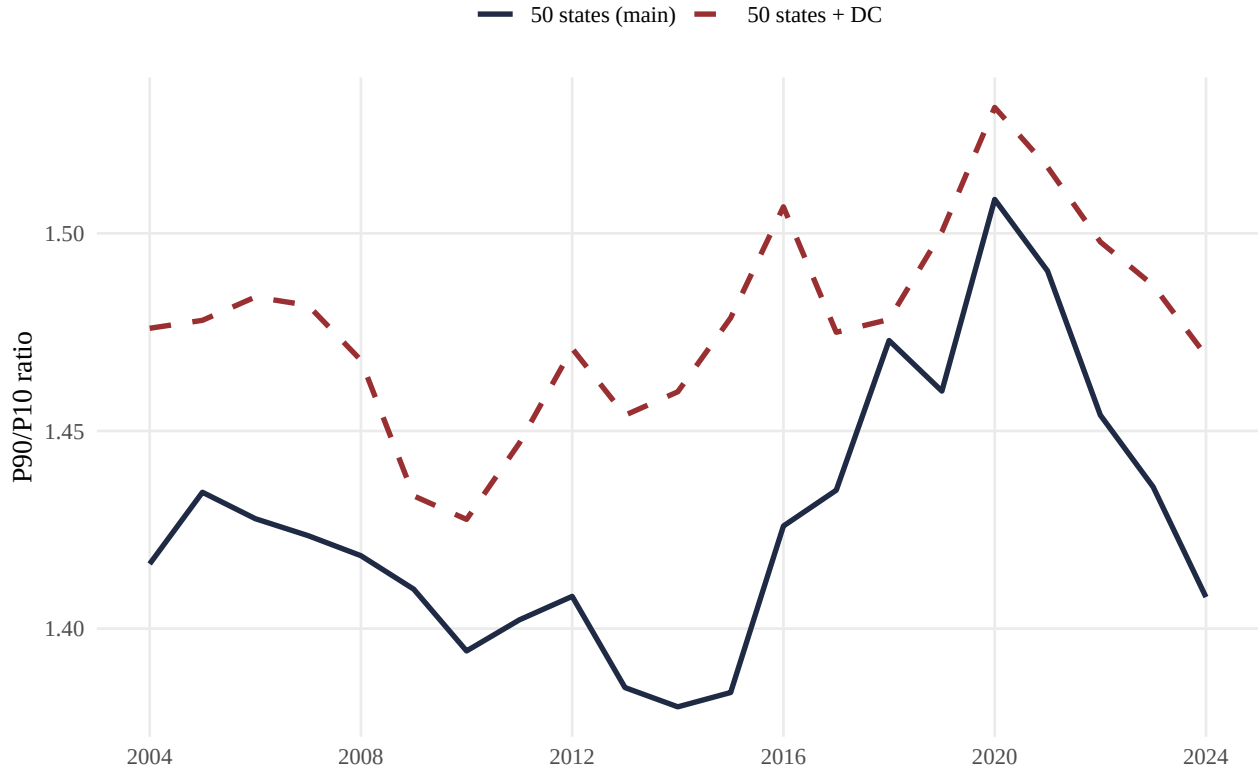


Figure 6: Robustness Check 2: P90/P10 with and without DC. DC’s structurally high federal-payroll average elevates the ratio in every year, but the time path is similar — our conclusion of headline stability does not depend on excluding DC.

```
bind_rows(
  ratio_full |> mutate(spec = "Main: 50 states"),
  ratio_no_covid |> mutate(spec = "Drop 2020-2021"),
  ratio_with_dc |> mutate(spec = "Include DC")
) |>
  filter(year %in% c(2004, 2014, 2024)) |>
  select(spec, year, ratio) |>
  mutate(ratio = round(ratio, 3)) |>
  pivot_wider(names_from = year, values_from = ratio) |>
  kable()
```

Table 6: Robustness comparison: P90/P10 across specifications, selected years.

spec	2004	2014	2024
Main: 50 states	1.416	1.38	1.408
Drop 2020-2021	1.416	1.38	1.408
Include DC	1.476	1.46	1.469

Check 3 (county-level cross-validation) is superseded by the full parallel county-level analysis presented in Section 5.

Section 5: County-level parallel analysis

The state-level analyses (Sections 3–4) find that the U.S. interstate wage gap has been roughly stable over 2004–2024 and that the modest underlying churn was concentrated in a handful of states. But state averages aggregate millions of workers across hundreds of counties whose own wages differ by 3× or more, so state-level stability is consistent with sharp divergence *between counties*. This section runs the same four analyses on a county panel to test that possibility.

Read the county panel

```
# Replace with the Dropbox link to qcew_county_panel.csv (ending in ?dl=1).
county_url <- "https://www.dropbox.com/scl/fi/tlnk7oms9y35fnp3kqynp/qcew_county_panel.csv?rlkey="

county_raw <- read_csv(
  county_url,
  show_col_types = FALSE,
  col_types = cols(area_fips = col_character(), .default = col_guess())
)

dim(county_raw)
```

```
[1] 67676      8
```

Filter the county panel

We restrict the county panel before analysis:

1. **Drop county-years outside the 50 states** (DC, PR, VI counties), identified by state FIPS (first two digits of `area_fips`).
2. **Drop county-years with zero or missing wage** (suppressed for confidentiality; small number of rows).
3. **Drop county-years with fewer than 500 average workers.** Counties this small are typically dominated by one or two employers, so the reported “average wage” reflects an idiosyncratic establishment rather than a functioning labor market. Removing them avoids letting outlier mini- counties drive the percentile measures. This drops about 2% of the panel; we report the count.

```
non_state_fips_2 <- c("11", "72", "78") # DC, PR, VI

county_panel <- county_raw |>
  mutate(state_fips_2 = substr(area_fips, 1, 2)) |>
  filter(!state_fips_2 %in% non_state_fips_2) |>
  filter(avg_annual_pay > 0, !is.na(avg_annual_pay)) |>
  filter(annual_avg_emplvl >= 500) |>
```

```

select(-state_fips_2)

# Add state and region, derive wage_real.
county_panel <- county_panel |>
mutate(
  state_fips_2 = substr(area_fips, 1, 2),
  region = case_when(
    state_fips_2 %in% c("09","23","25","33","34","36","42","44","50")
      ~ "Northeast",
    state_fips_2 %in% c("17","18","19","20","26","27","29","31","38","39",
      "46","55") ~ "Midwest",
    state_fips_2 %in% c("01","05","10","12","13","21","22","24","28","37",
      "40","45","47","48","51","54") ~ "South",
    state_fips_2 %in% c("02","04","06","08","15","16","30","32","35","41",
      "49","53","56") ~ "West"
  ),
  wage_real = avg_annual_pay * (cpi_2024 / cpi_u)
) |>
select(-state_fips_2)

```

```

tibble(
  step = c(
    "Raw county panel",
    "After dropping DC, PR, VI counties",
    "After dropping suppressed / zero-wage",
    "After dropping county-years with < 500 workers"
  ),
  rows = c(
    nrow(county_raw),
    nrow(county_raw |>
      filter(!substr(area_fips, 1, 2) %in% non_state_fips_2)),
    nrow(county_raw |>
      filter(!substr(area_fips, 1, 2) %in% non_state_fips_2) |>
      filter(avg_annual_pay > 0, !is.na(avg_annual_pay))),
    nrow(county_panel)
  )
) |>
kable()

```

step	rows
Raw county panel	67676
After dropping DC, PR, VI counties	65954
After dropping suppressed / zero-wage	65946
After dropping county-years with < 500 workers	64497

The analytical county panel has 64,497 county-year observations covering 3,107 distinct counties.

County Analysis 1 — Dispersion ratios over time (Figure C1)

```
county_percentiles_year <- county_panel |>
  group_by(year) |>
  summarise(
    p10 = quantile(wage_real, 0.10),
    p25 = quantile(wage_real, 0.25),
    p50 = quantile(wage_real, 0.50),
    p75 = quantile(wage_real, 0.75),
    p90 = quantile(wage_real, 0.90),
    .groups = "drop"
  ) |>
  mutate(
    `P90/P10` = p90 / p10,
    `P90/P50` = p90 / p50,
    `P50/P10` = p50 / p10
  )

county_ratios_long <- county_percentiles_year |>
  select(year, `P90/P10`, `P90/P50`, `P50/P10`) |>
  pivot_longer(-year, names_to = "ratio", values_to = "value") |>
  mutate(ratio = factor(ratio, levels = c("P90/P10", "P90/P50", "P50/P10")))

ggplot(county_ratios_long, aes(x = year, y = value,
                              color = ratio, linewidth = ratio)) +
  annotate("rect", xmin = 2008, xmax = 2009, ymin = -Inf, ymax = Inf,
          fill = "gray85", alpha = 0.5) +
  annotate("rect", xmin = 2020, xmax = 2021, ymin = -Inf, ymax = Inf,
          fill = "gray85", alpha = 0.5) +
  geom_line() +
  scale_color_manual(values = c("P90/P10" = "#1f2a44",
                                "P90/P50" = "#992f31",
                                "P50/P10" = "#7c8a9b")) +
  scale_linewidth_manual(values = c("P90/P10" = 1.4,
                                    "P90/P50" = 0.8,
                                    "P50/P10" = 0.8)) +
  scale_x_continuous(breaks = seq(2004, 2024, 4)) +
  guides(linewidth = "none") +
  labs(x = NULL, y = "Ratio (across counties)", color = NULL) +
  theme(
    text = element_text(family = "serif"),
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "gray92"),
    legend.position = "inside",
    legend.position.inside = c(0.15, 0.85),
    legend.background = element_rect(fill = "white", color = NA)
  )
```

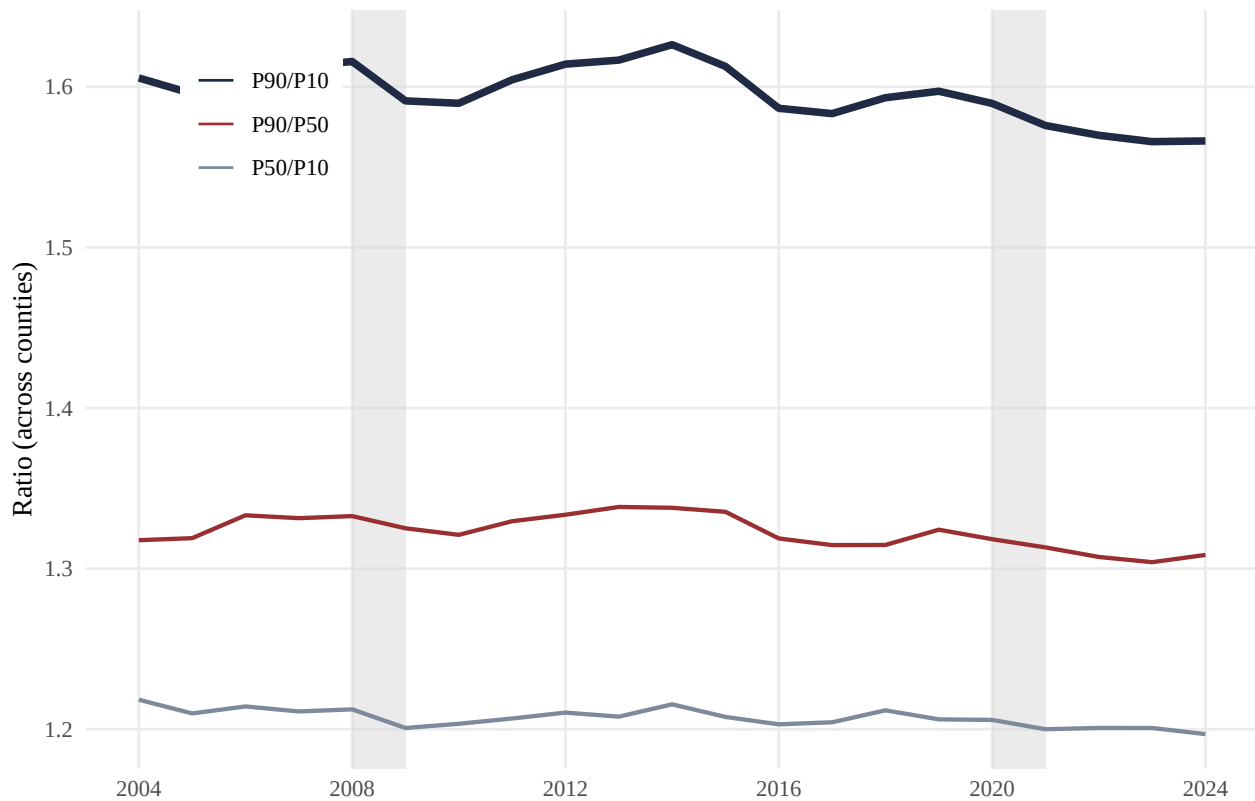


Figure 7: Figure C1: Dispersion ratios across all U.S. counties (50 states), 2004–2024.

County Analysis 2 — Indexed percentile trajectories (Figure C2)

```
county_indexed_long <- county_percentiles_year |>
  select(year, p10, p25, p50, p75, p90) |>
  pivot_longer(-year, names_to = "percentile", values_to = "value") |>
  group_by(percentile) |>
  mutate(index = value / value[year == 2004] * 100) |>
  ungroup() |>
  mutate(percentile = factor(
    percentile,
    levels = c("p90", "p75", "p50", "p25", "p10"),
    labels = c("P90", "P75", "P50", "P25", "P10")
  ))

county_end_labels <- county_indexed_long |> filter(year == 2024)

ggplot(county_indexed_long, aes(x = year, y = index,
                                color = percentile, linewidth = percentile)) +
  geom_hline(yintercept = 100, color = "gray70", linetype = "dashed",
             linewidth = 0.4) +
  geom_line() +
```

```

geom_text(data = county_end_labels, aes(label = percentile),
          hjust = -0.2, size = 3.2, family = "serif",
          show.legend = FALSE) +
scale_color_manual(values = c("P90" = "#992f31", "P75" = "#1f2a44",
                              "P50" = "#4a5a7a", "P25" = "#7c8a9b",
                              "P10" = "#b1bdc3")) +
scale_linewidth_manual(values = c("P90" = 1.4, "P75" = 0.8,
                                  "P50" = 0.8, "P25" = 0.8, "P10" = 0.8)) +
scale_x_continuous(breaks = seq(2004, 2024, 4),
                   limits = c(2004, 2025.5)) +
guides(linewidth = "none", color = "none") +
labs(x = NULL, y = "Indexed value (2004 = 100)") +
theme(
  text = element_text(family = "serif"),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray92")
)

```

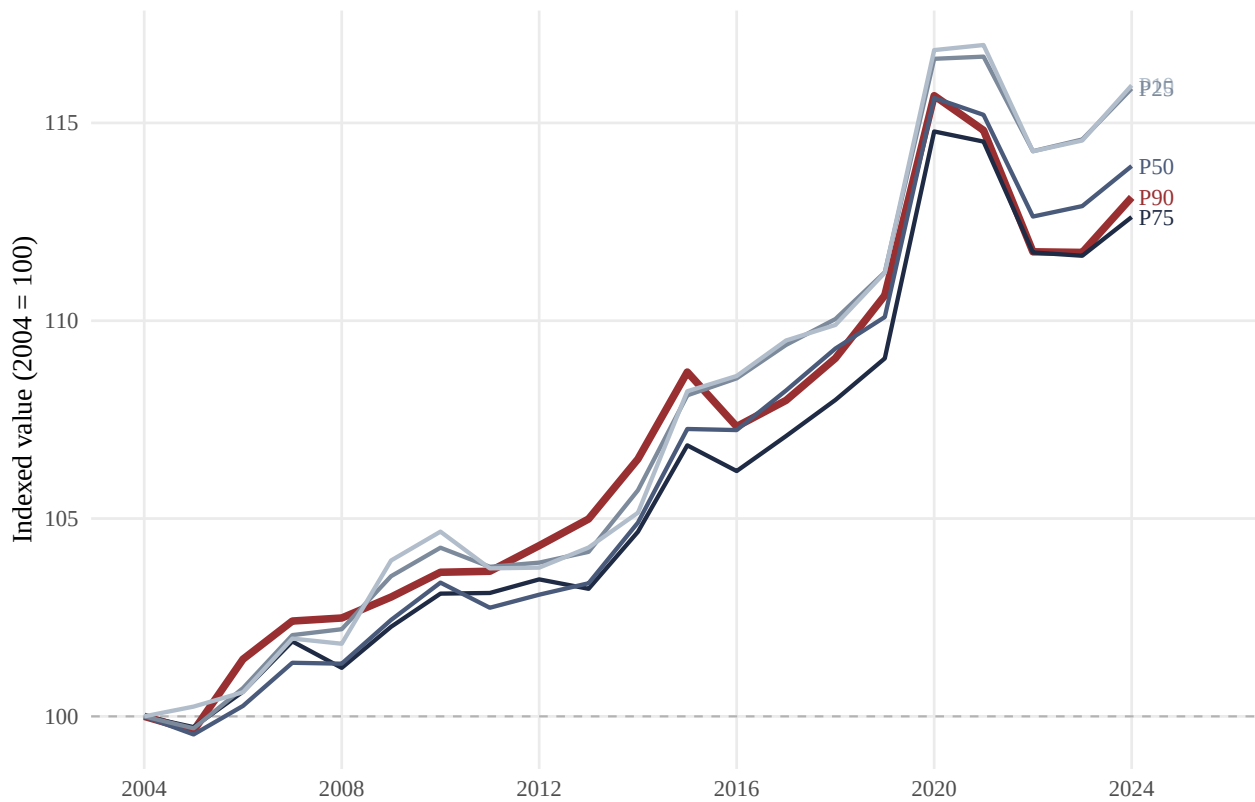


Figure 8: Figure C2: Across-county percentiles indexed to 100 in 2004.

County Analysis 3 — Growth-incidence scatter (Figure C3)

```
# Restrict to counties present in BOTH 2004 and 2024 (so growth is defined).
county_balanced <- county_panel |>
  filter(year %in% c(2004, 2024)) |>
  group_by(area_fips) |>
  filter(n() == 2) |>
  ungroup()

county_growth <- county_balanced |>
  select(area_fips, area_title, region, year, wage_real) |>
  pivot_wider(names_from = year, values_from = wage_real,
              names_prefix = "wage_real_") |>
  mutate(pct_growth_2004_2024 = (wage_real_2024 / wage_real_2004) - 1) |>
  filter(!is.na(pct_growth_2004_2024))

top_10_counties <- county_growth |> slice_max(pct_growth_2004_2024, n = 10)
bottom_10_counties <- county_growth |> slice_min(pct_growth_2004_2024, n = 10)
to_label_counties <- bind_rows(top_10_counties, bottom_10_counties)
```

```
ggplot(county_growth,
       aes(x = wage_real_2004 / 1000, y = pct_growth_2004_2024)) +
  geom_hline(yintercept = 0, color = "gray70", linetype = "dotted",
            linewidth = 0.4) +
  geom_smooth(method = "loess", se = FALSE,
             color = "gray40", linewidth = 0.6, linetype = "dashed",
             formula = y ~ x) +
  geom_point(aes(color = region), size = 1.0, alpha = 0.5) +
  geom_text_repel(
    data = to_label_counties,
    aes(label = area_title),
    size = 2.6, family = "serif",
    box.padding = 0.5, max.overlaps = Inf, seed = 42,
    min.segment.length = 0
  ) +
  scale_color_manual(values = c("Northeast" = "#1f2a44",
                                "Midwest" = "#4a7a4a",
                                "South" = "#992f31",
                                "West" = "#b58b3a")) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  scale_x_continuous(labels = scales::label_dollar(suffix = "K")) +
  labs(
    x = "2004 real wage (thousands of 2024 dollars)",
    y = "Cumulative real wage growth, 2004 → 2024",
    color = NULL
  ) +
```



```
theme(
  text = element_text(family = "serif"),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray92"),
  legend.position = "top"
)
```

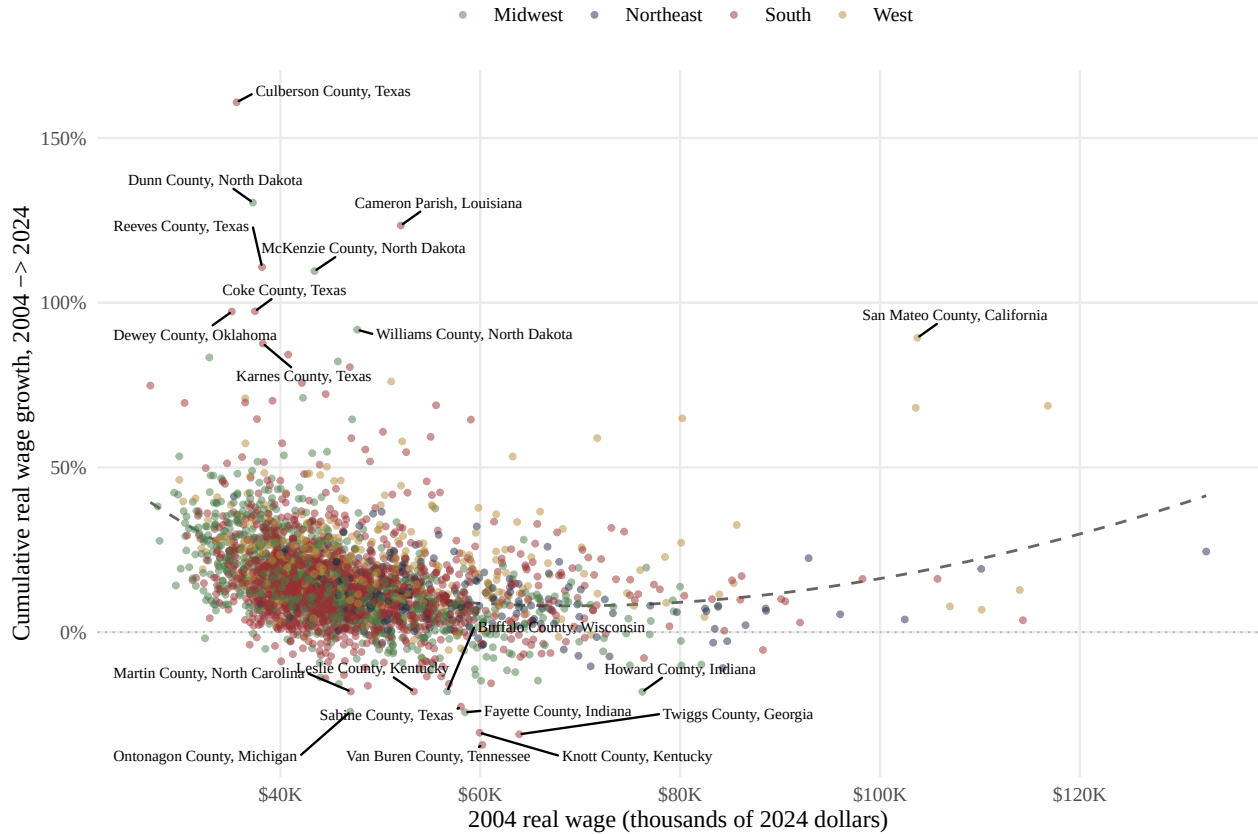


Figure 9: Figure C3: 2004 real wage vs. 2004–2024 real wage growth, U.S. counties. Top 10 fastest- and bottom 10 slowest-growing labeled. Counties present in both 2004 and 2024 only.

County Analysis 4 — Concentration test (Figure C4)

```
compute_county_p90p10 <- function(df) {
  df |> group_by(year) |>
    summarise(p10 = quantile(wage_real, 0.10),
              p90 = quantile(wage_real, 0.90),
              ratio = p90 / p10, .groups = "drop")
}

top_10_county_fips <- top_10_counties$area_fips
bot_10_county_fips <- bottom_10_counties$area_fips
```

```

county_ratio_full <- compute_county_p90p10(county_panel) |>
  mutate(series = "Full sample (all counties)")
county_ratio_no_top10 <- county_panel |>
  filter(!area_fips %in% top_10_county_fips) |>
  compute_county_p90p10() |>
  mutate(series = "Drop 10 fastest-growth counties")
county_ratio_no_bot10 <- county_panel |>
  filter(!area_fips %in% bot_10_county_fips) |>
  compute_county_p90p10() |>
  mutate(series = "Drop 10 slowest-growth counties")

county_ratios_concentration <- bind_rows(
  county_ratio_full, county_ratio_no_top10, county_ratio_no_bot10
) |>
  mutate(series = factor(series, levels = c(
    "Full sample (all counties)",
    "Drop 10 fastest-growth counties",
    "Drop 10 slowest-growth counties"
  )))

```

```

ggplot(county_ratios_concentration,
  aes(x = year, y = ratio, color = series,
      linetype = series, linewidth = series)) +
  geom_line() +
  scale_color_manual(values = c(
    "Full sample (all counties)" = "#1f2a44",
    "Drop 10 fastest-growth counties" = "#992f31",
    "Drop 10 slowest-growth counties" = "#7c8a9b"
  )) +
  scale_linetype_manual(values = c(
    "Full sample (all counties)" = "solid",
    "Drop 10 fastest-growth counties" = "dashed",
    "Drop 10 slowest-growth counties" = "dotdash"
  )) +
  scale_linewidth_manual(values = c(
    "Full sample (all counties)" = 1.3,
    "Drop 10 fastest-growth counties" = 0.9,
    "Drop 10 slowest-growth counties" = 0.9
  )) +
  scale_x_continuous(breaks = seq(2004, 2024, 4)) +
  guides(linewidth = "none",
    linetype = guide_legend(override.aes = list(linewidth = 0.8))) +
  labs(x = NULL, y = "P90/P10 ratio (across counties)",
    color = NULL, linetype = NULL) +
  theme(
    text = element_text(family = "serif"),
    panel.grid.minor = element_blank(),

```

```

panel.grid.major = element_line(color = "gray92"),
legend.position = "top",
legend.text = element_text(size = 9)
)

```

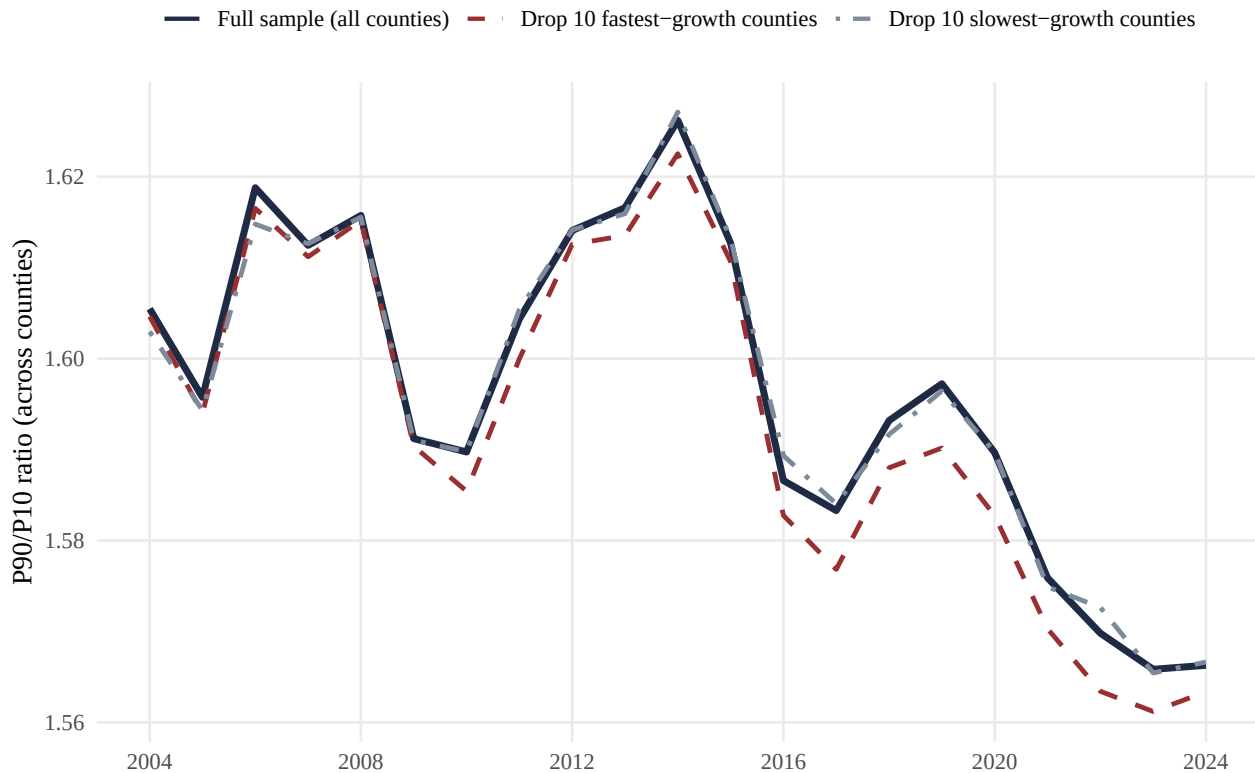


Figure 10: Figure C4: P90/P10 across counties, recomputed three ways. Dropping the 10 fastest-growth counties barely moves the line — the divergence is broad, not concentrated in a handful of superstar counties.

Section 6: State vs. county — the bridge figure

```

bridge_state <- ratio_full |>
  rename(state_ratio = ratio) |>
  select(year, state_ratio)
bridge_county <- county_ratio_full |>
  rename(county_ratio = ratio) |>
  select(year, county_ratio)

bridge <- inner_join(bridge_state, bridge_county, by = "year") |>
  mutate(
    State = state_ratio / state_ratio[year == 2004] * 100,
    County = county_ratio / county_ratio[year == 2004] * 100
  )

```

```

) |>
select(year, State, County) |>
pivot_longer(-year, names_to = "level", values_to = "index") |>
mutate(level = factor(level, levels = c("County", "State")))

ggplot(bridge, aes(x = year, y = index, color = level, linewidth = level)) +
  geom_hline(yintercept = 100, color = "gray70", linetype = "dashed",
            linewidth = 0.4) +
  geom_line() +
  scale_color_manual(values = c("County" = "#992f31", "State" = "#1f2a44")) +
  scale_linewidth_manual(values = c("County" = 1.4, "State" = 1.4)) +
  scale_x_continuous(breaks = seq(2004, 2024, 4)) +
  labs(x = NULL, y = "P90/P10 ratio, indexed to 100 in 2004",
       color = NULL) +
  guides(linewidth = "none") +
  theme(
    text = element_text(family = "serif"),
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "gray92"),
    legend.position = "top"
  )

```

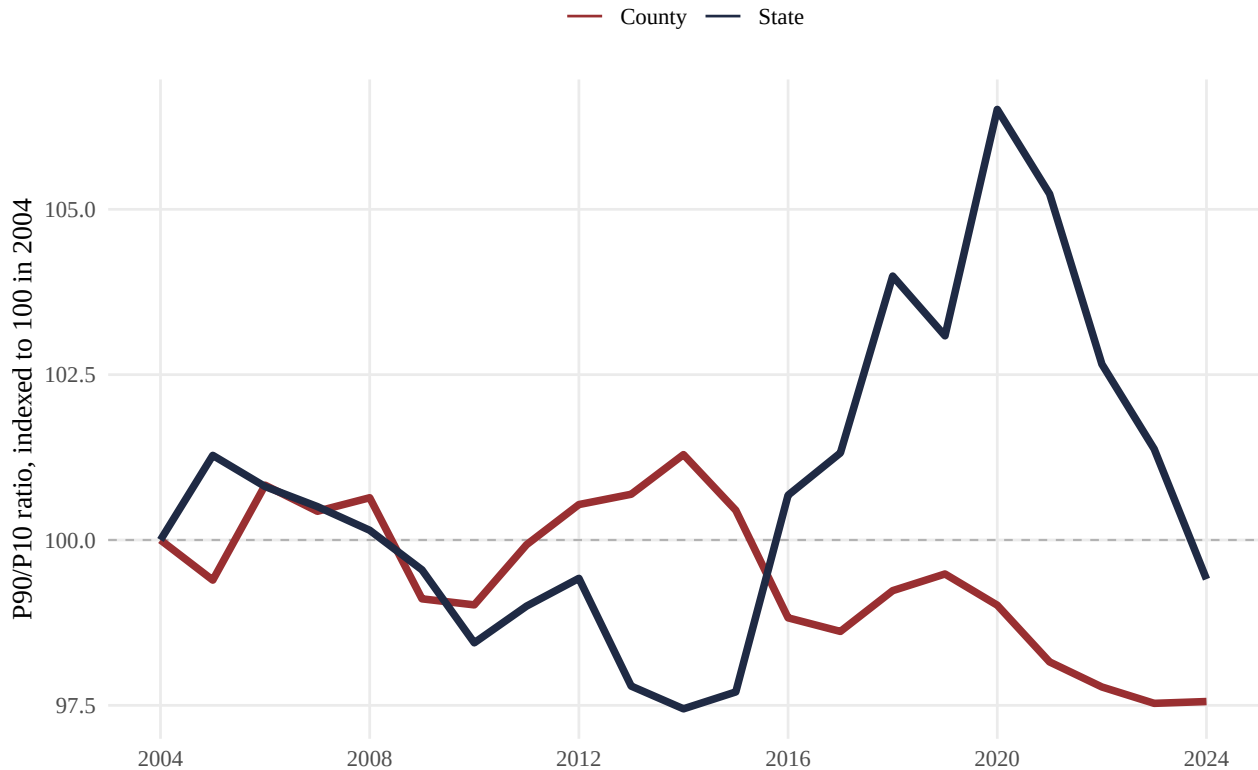


Figure 11: Figure B: The headline P90/P10 ratio across states (blue) vs. across counties (red), each indexed to 100 in 2004. Both series end near their 2004 baseline. The state series shows the COVID composition spike; the county series is smoother and ends slightly below 100. Geographic dispersion at the percentile level was stable across scales.

Interpretation. The two lines tell the project’s full thesis, but it is not the one we expected before running the data. At every geographic scale we examined — 50 states and 3,000+ counties — the headline P90/P10 gap was roughly stable over 2004–2024 (state: 1.42 → 1.41; county: 1.61 → 1.57). What changed dramatically was *who occupies the top and bottom of those distributions*. County-level real wage growth ranged from –34% to +161%; eight of the ten fastest-growing counties were Permian Basin or Bakken shale-oil counties; the slowest-growing were in rural Appalachia and the Rust Belt. **Aggregate U.S. geographic wage inequality measured by P90/P10 looks essentially as it did in 2004 — but the geography of who sits where has been rewritten by the shale boom, manufacturing decline, and a handful of coastal tech counties.**

This finding pushes back on two common narratives. It is not the case that a few coastal cities “pulled away” from the rest of the country at this dispersion measure — San Mateo County is the only non-energy county in the top 10. And it is not the case that states or counties broadly diverged at the percentile level. The real geographic wage story of the last two decades was industry-specific churn within a stable distribution of relative wages.